Geant4 simulation for QuantumCosmic detector.

Carolina Filgueira Rama

Detector description

The QuantumCosmic detector is designed with the purpose of measuring the cosmic rays impacting in a certain quantum processor unit (QPU), in this case for the QMIO quantum computer, at CESGA (Santiago de Compostela).

The QPU is estimated to measure about 22×22 mm^2, which is a relatively small area, and it is incorporated in a cryogenic tank, since it operates with superconducting qubits. To measure an impact of a cosmic ray in this unit, the idea is to set two planes of scintillating bars behind the cryogenic unit, as near to the QPU as possible.

The detector consists of an upper plane (plane 1) with 60 scintillating bars of $15 \times 15 \times 900$ mm^3, having an approximate area of 900×900 mm^2 (not perfectly square since we have some coating material covering the bars), and a lower plane (plane 2) with 90 scintillating bars of $15 \times 15 \times 1500$ mm^3. A representation can be found in Figure 1.

The reconstruction principle relies on determining the hit position of a cosmic ray in both planes. From the two reconstructed points, one can derive the equation of the straight line corresponding to the particle trajectory and extrapolate whether the track intersects the QPU. The impact positions within each bar are inferred from the difference in arrival times of scintillation photons at the two ends of the bar, assuming the photon propagation velocity in the medium is known.

The critical question for the detector design is whether the timing resolution of the readout system is sufficient to achieve the required spatial precision. Electronic limitations and experimental constraints may prevent the time difference from being measured with enough accuracy. To quantify these effects, a detailed Geant4 simulation [1] was developed. The goal of this simulation is to reproduce the detector response and to study how uncertainties in the reconstructed hit positions depend on the assumed timing resolution.

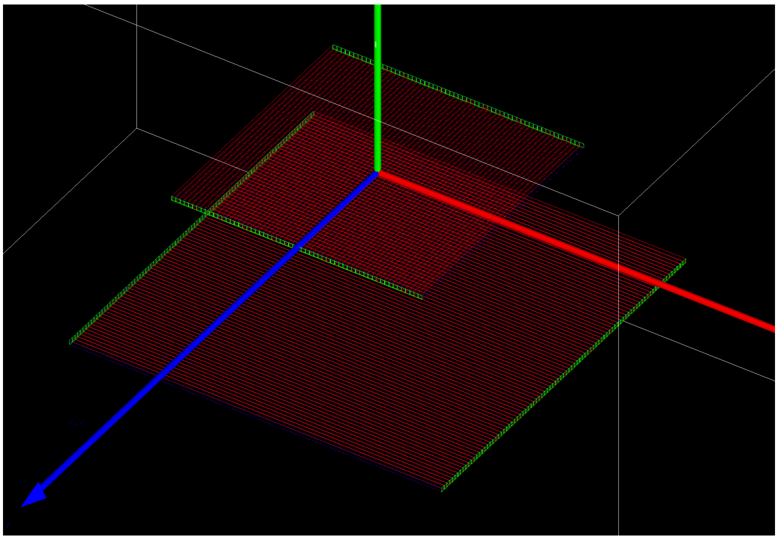


Figure 1: QuantumCosmic detector simulated with Geant4.

Code structure

To simulate the behaviour of the detector, three main points where covered: the implementation of the cosmic rays, the design of the detector and the measured quantities for a later analysis. We will provide here some generic comments to make easier to follow the later analysis.

Cosmic rays implementation: CRY software

To generate the cosmic rays in the simulation, the CRY software library [2] was used. This software was added to the Geant4 simulation so that it can be used directly each time the simulation is run. This software provides the distribution of cosmic rays at sea level (around Santiago's level) and uses a random generator to give the initial momentum of the particles. An analysis of this quantities is done to check the validity of this software.

Detector implementation

The construction of the detector is perhaps the most important part in any Geant4 simulation. In order to recreate the behaviour of the particles, the materials must be clearly defined. In our case, the materials' properties are generic, since the specific components to be used in the real detector have not yet been determined.

As can be seen in Figure 1, plane 1's scintillating bars are aligned with the z-axis, while plane 2's bars are aligned perpendicular to them, in this case with the x-axis. These planes have been arbitrarily set to $y_1 = 0\ mm$ and $y_2 = -500\ mm$. Note that the Geant4 coordinate system has a different orientation to the standard one, with the vertical axis being y.

Although not visualized in the figure, the scintillating bars are covered with a $0,1\ mm$ of thick polyethylene coating, to avoid light losses. With this thickness, the two planes have dimensions of $900 \times 912\ mm^2$ and $1500 \times 1518\ mm^2$.

The QPU has also been declared as part of the detector, defined as an square of $22 \times 22\ mm2$ aligned with the y direction (oriented vertically), at a height of $y_{qpu} = 500\ mm$.

The construction of the detector depends on the simulation to be carried out. Currently, we have only implemented a simulation with the scintillation process parametrized externally. However, Geant4 provides an option to generate the scintillation process internally by providing the optical properties of the materials. This is a very detailed implementation with several subtleties. This simulation mode has not yet been implemented correctly, so for the time being, we will only cover the parametrized mode.

Measured quantities

As stated at the outset, the measurement to be recorded in the real detector is the difference in time at each end of the scintillating bar. To study the detector's performance, we must first analyse these quantities (the times measured at each end of the bar) to determine the impact points in each plane required for the projection of the track and recovery of the impact point at the QPU level.

Several magnitudes will be generated in the code and saved in an external file to perform this analysis.

As we have not specified the optical properties required for Geant4 to generate scintillating photons internally, we must parametrize their behaviour. To do this, we first recover the impact of a cosmic ray on a scintillating bar and assign a group velocity in the scintillating material of $v_g = c/n$. Here, n is the refractive index of the material. For our purposes, we set $n = 1.58$. We can then compute the time it takes a photon to reach each end of the bar (t_A and t_B).

Note that when computing these times, we only consider the coordinate along which the bar is aligned in each case: the z coordinate for plane 1 and the x coordinate for plane 2.

For the later analysis we will save several magnitudes that will be needed for the analysis, such as the ID of the impacted bar, the ID of the plane, both the local coordinates of the impact with respect to the bar, and the global coordinates with respect to the whole detector, among others.

Geant4 handles the detection of cosmic ray impacts on scintillating bars using an internal structure called a 'hit'. These hits are stored each time a particle deposits energy in the material it is passing through. If this material is set as a sensitive detector (another Geant4 structure that handles the detection of particles within a material), the hits are stored. Bearing this in mind, we have chosen to save both the first and latest hits of the cosmic rays in the bar (obtained with respect to the global time of the simulation). This means that for each event in the simulation (i.e. each time a cosmic ray is sent), we have several impacts (hits) saved for each bar in each plane.

Analysis of the simulated data

The generated data explained in the previous section needs to be analysed to study the validity of both the simulation and the detector setup.

Cosmic rays

We start by analysing the quality of the cosmic rays generated with the external CRY software. As it is well known, the cosmic rays at sea level follow an angular distribution of $f(\theta) = cos^2\theta$ [3]. To assess the quality of the cosmic rays, we will convert the generated momenta to spherical coordinates. It should be noted here that the definition of spherical coordinates must be adapted for use with the Geant4 coordinate system, that is:

$$\theta = arccos\frac{z}{\sqrt{x^2+y^2+z^2}} \qquad \varphi = arctan\frac{x}{z} \tag{1}$$

With respect to the canonical coordinate system, the Geant4 coordinates can be obtained by making the following changes:

$$x \rightarrow z$$
$$y \rightarrow x$$
$$z \rightarrow y$$

With these definitions clear, we studied one million cosmic rays sent through a plane half the size of the simulation's world volume (the volume within which the simulation is defined) above the scintillating planes.

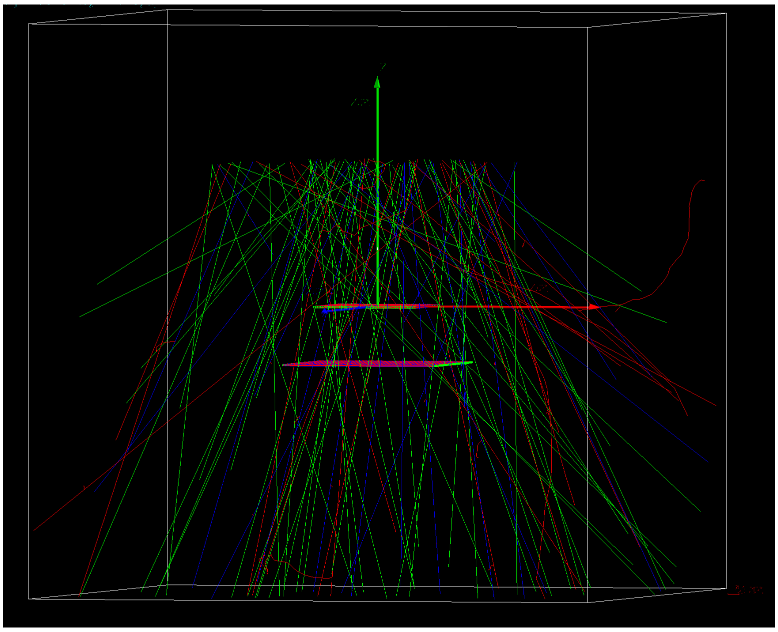


Figure 2: Random distribution of cosmic rays to study its quality.

Using this configuration, we recovered the angular distribution of their momenta to obtain the following result:

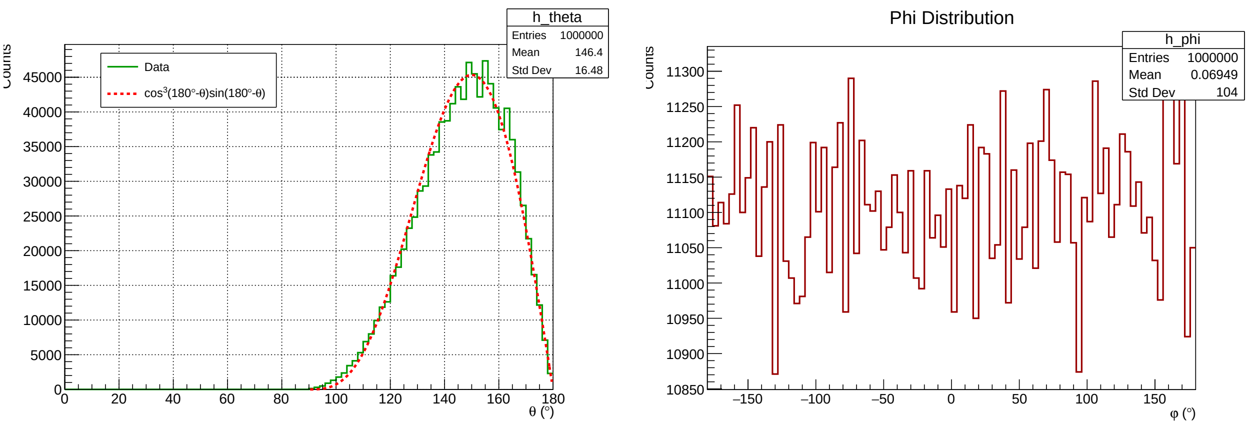


Figure 3: Angular momenta distribution of the generated cosmic rays with CRY.

We can see that the θ distribution follows a theoretical distribution of $g(\theta)d\theta = cos^3\theta sin\theta d\theta$, which is the expected distribution, $f(\theta)$ with the differential element for spherical coordinates and an extra $cos\theta$ factor that comes from considering the effective area of the detection. For the φ distribution we observe an approximately uniform distribution (note the range on the y axis).

We can also plot the positions generated by the CRY software within the expected area to see if there is uniformity:

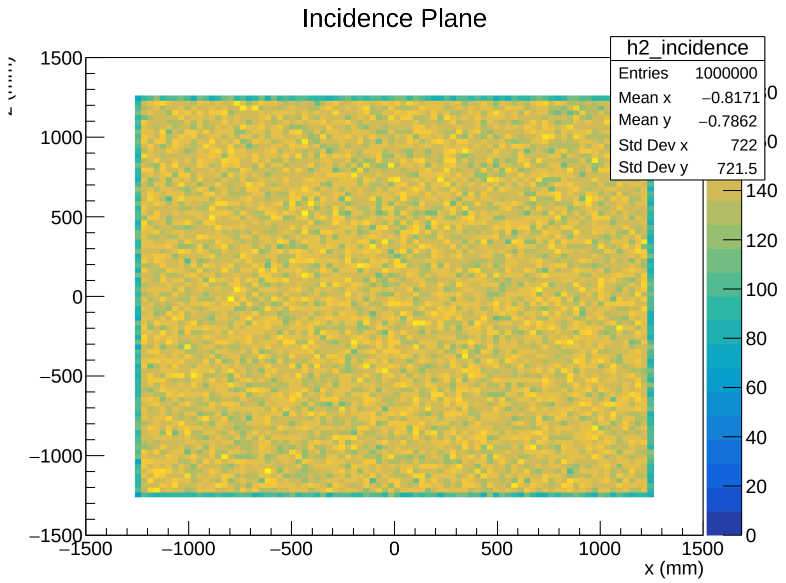


Figure 4: Distribution of the generated cosmic rays using CRY along its generation plane.

As expected, we see a uniform distribution across the plane.

In addition to recovering the momentum and position distributions, we can present the particle distribution of the generated data, to see if it is as expected:

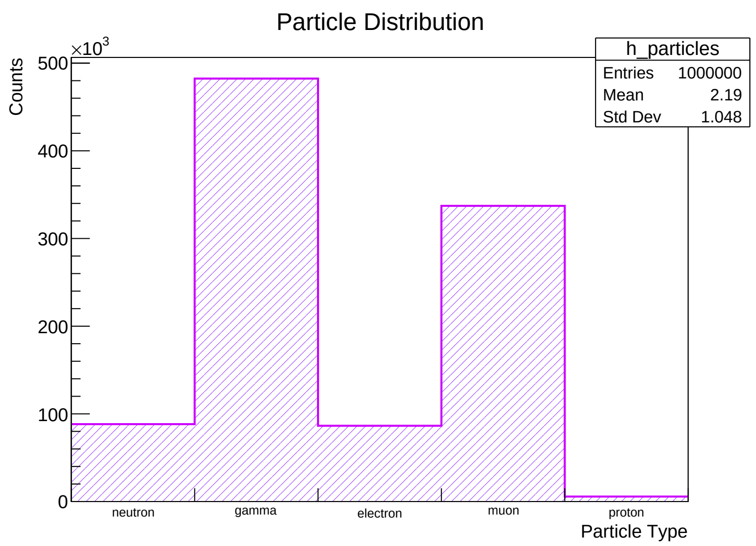


Figure 5: Particle distribution of CRY generated cosmic rays at sea level.

Lastly, we have also examined how the energy of the cosmic rays is distributed among all particles, muons and electrons.

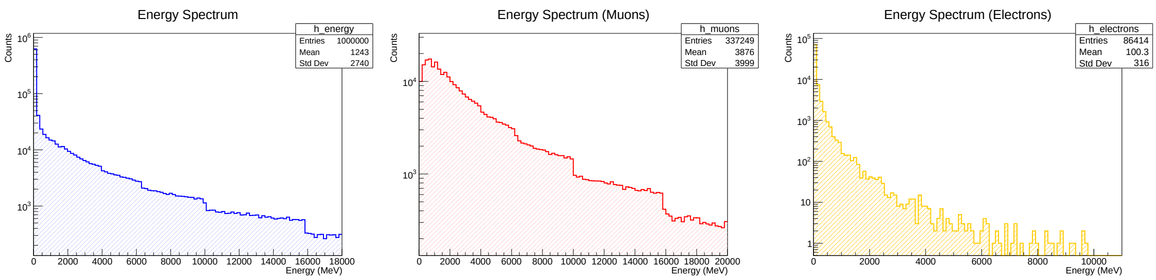


Figure 6: Energy disrtibution of the cosmic rays generated with CRY for different sets of particles.

We observe some discontinuities in the energy spectra that have already been identified in previous studies of the cosmic ray generation software [4]. Overall, we conclude that the CRY generation software is valid for our purposes.

Detector resolution

As we explained earlier, we are now going to study now in detail the time resolution required for the detector to obtain a precise reconstruction of the impact point in the QPU.

For this study, all of the generated cosmic rays will come from the middle point in the QPU, that is, with respect to the global reference frame $(0, 500, 0)$ mm. As a visual aid we provide the simulation scheme for this case in Figure 7.

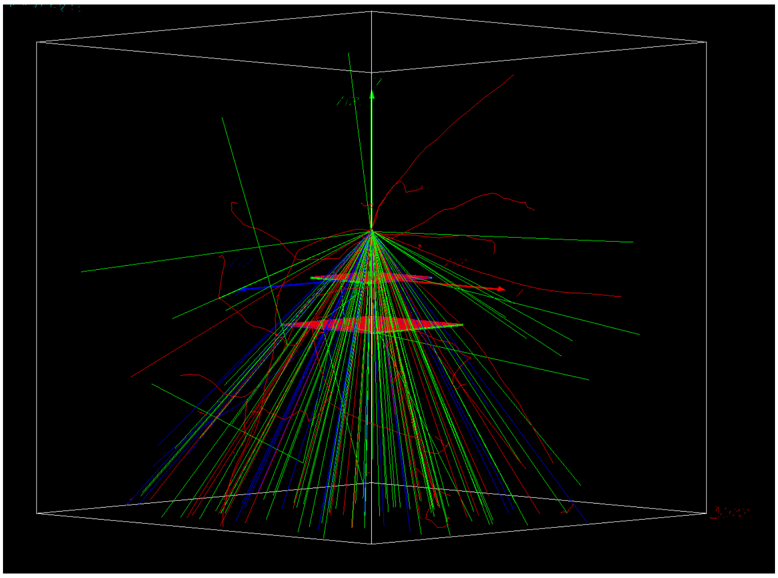
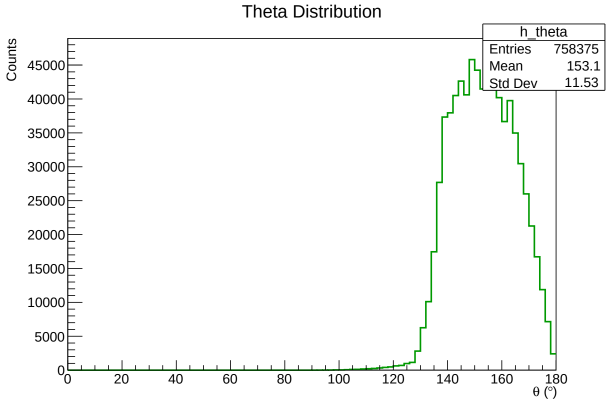


Figure 7: Generated cosmic rays at the QPU center point.

We can start by analysing the angular distribution of the cosmic rays to see if anything is different from what we would expect:



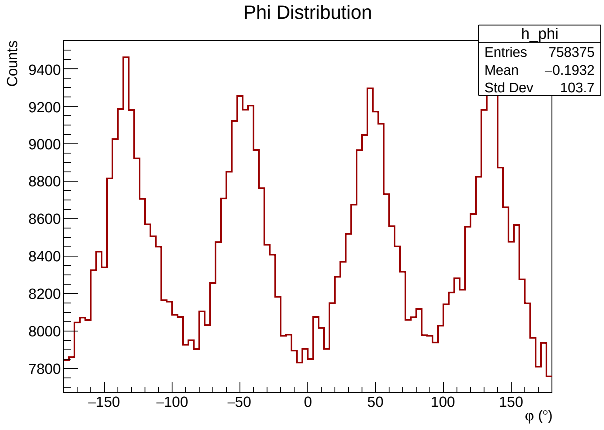


Figure 8: Angular momenta distribution of the generated cosmic rays with CRY with the QPU as starting point.

We can observe some differences with respect to the distributions shown in Figure 3, the φ coordinate appears to exhibit a periodic behaviour, and the θ distribution is less smooth than in the previous case.

We can also plot the cosmic ray particle distribution measured at the bars:

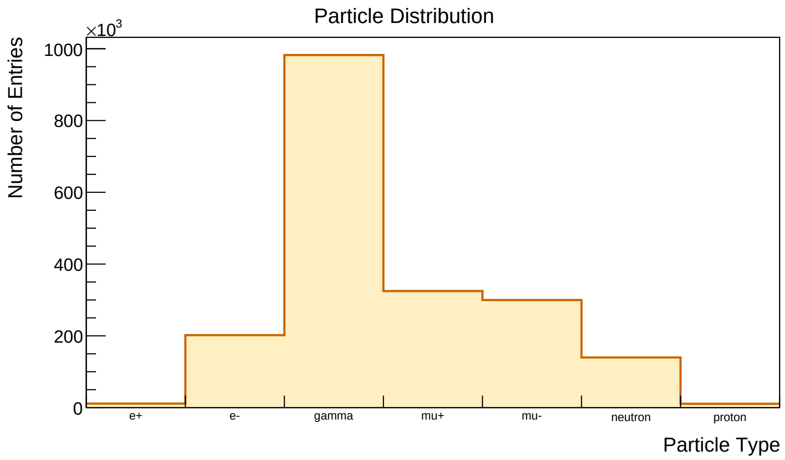


Figure 9: Particle distribution of the cosmic rays measured in the scintillating bars.

We observe a distribution similar to that presented in Figure 5, except that here the muons are separated into two bins according to their charge. We can now begin analysing the distribution of impacts in relation to the detector and the bars. To do so, we used two different reference frames: a global frame of reference with respect to the general detector, and a local frame of reference centred on the midpoint of each bar.

We recall here that we have saved both the impact points for the first and last hits, thus for the following representations we have used the midpoint between the two. It is interesting to note here that only 8.97% of the muons had different impact points at the beginning and at the end, whereas the value becomes 20.60% when considering all of the particles. This is consistent with the behaviour we would expect from muons, given that they have the smallest stopping power of all the particles.

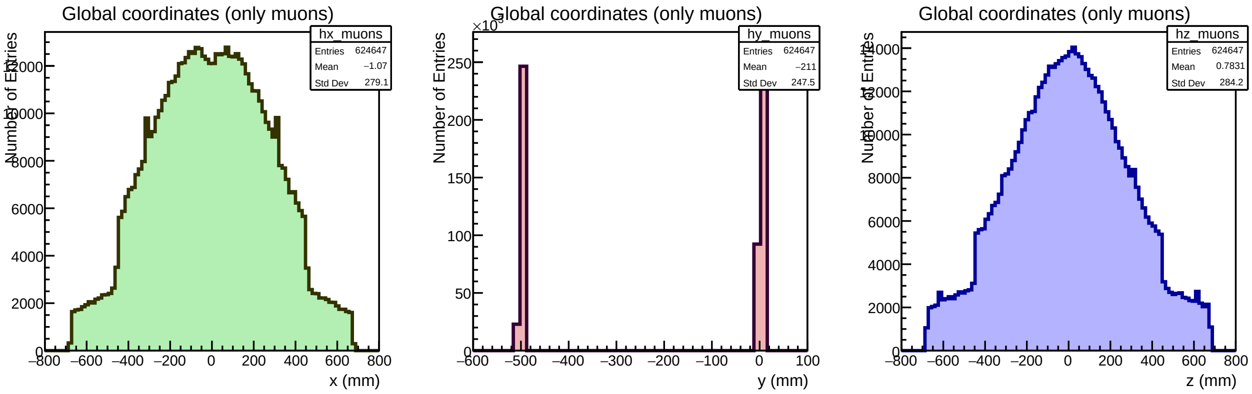


Figure 10: Distribution of the impact points with respect to the global reference frame for muons only.

As can be seen in the graphs, there is a step in both x and z coordinates. This is because the plane 2 is larger than the plane 1. We can also observe that these planes are located at $y_1 = 0\ mm$ and $y_2 = -500\ mm$ from the y coordinate distribution.

Here, we have considered only the muons, because they are the particles that are expected to be measured most frequently at our altitude level. From now on, we will be providing the information only for muons.

To provide an alternative perspective, we have also saved the impact points of the cosmic rays in relation to the local reference frame. For each bar, the reference frame is located at its centre, as explained earlier.

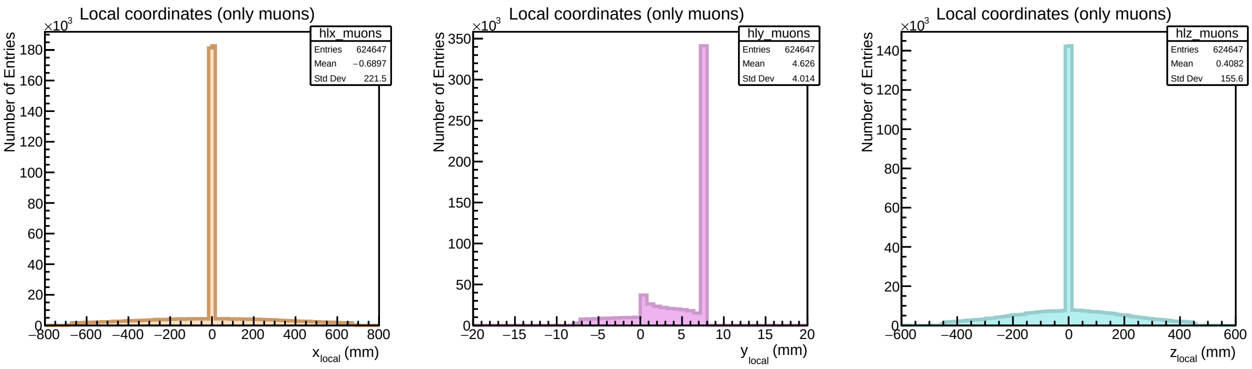


Figure 11: Distribution of the impact points with respect to the local reference frame for muons only.

Here we can observe differences from the previous graphs. The first thing to notice is that most of the impacts are at the highest value with respect to the y coordinate, as one could have guessed. We notice also that in x and z histograms there is a superposition of two distributions, one spread along the bar length and another one that resembles a Dirac delta function. This is because we are plotting both values from plane 1 and 2. For example, for plane 1 values, the z coordinate is expected to have values covering all the length of the bar, while the x coordinate is expected to have values centered at 0, with a width of 15 mm as maximum, and the analogous happens for the plane 2 values.

We will also provide colour maps of the impacts along each plane to check whether they correspond to the expected central distribution (see Figure 7). If we represent the impact points with respect to the global reference frame:

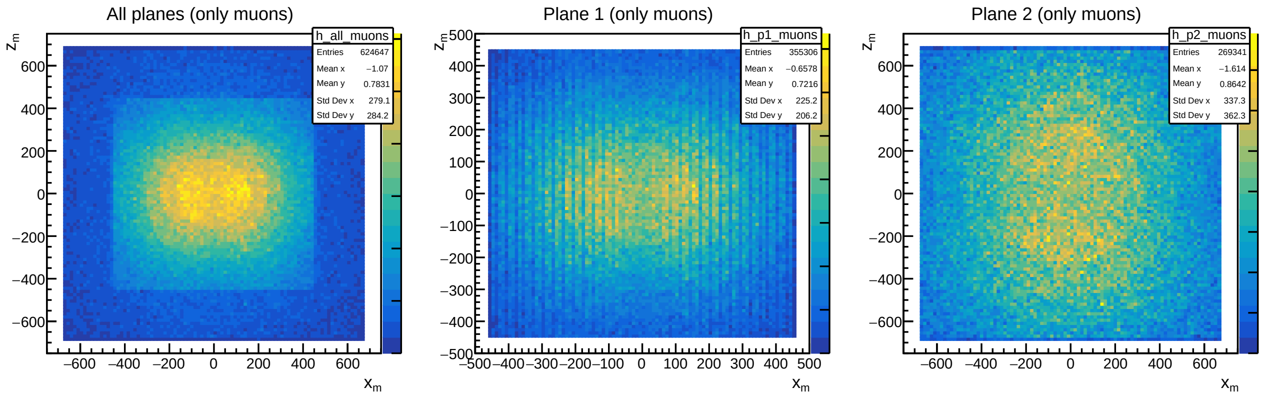


Figure 12: 2D histograms of the impact points in the scintillating bars for the global reference frame.

And the same impact points with respect to the local reference frame:

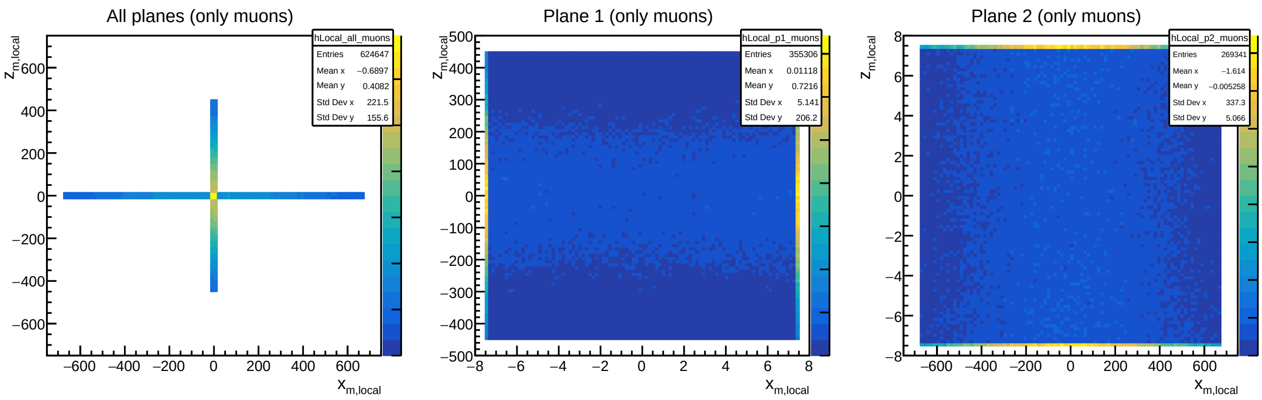


Figure 13: 2D histograms of the impact points in the scintillating bars for the local reference frame.

From this two dimensional histograms we can observe some things. Fistly, we can see that in plane 1 some of the bars are more prevalent others, which is unexpected. In contrast, in plane 2 we do see the expected behaviour of a central distribution of impacts. We can also notice, when observing the hits in the local reference frame that the cosmic rays tendx to impact at the borders of the bars, which, in principle, should not happen.

We can now study the behaviour of the times measured at each of the endpoints of the bar, t_A and t_B. These times are obtained by parameterising the behaviour of the photons in a scintillating bar, as previously explained. We will present the latest hit values here, since there is no noticeable difference between the distributions.

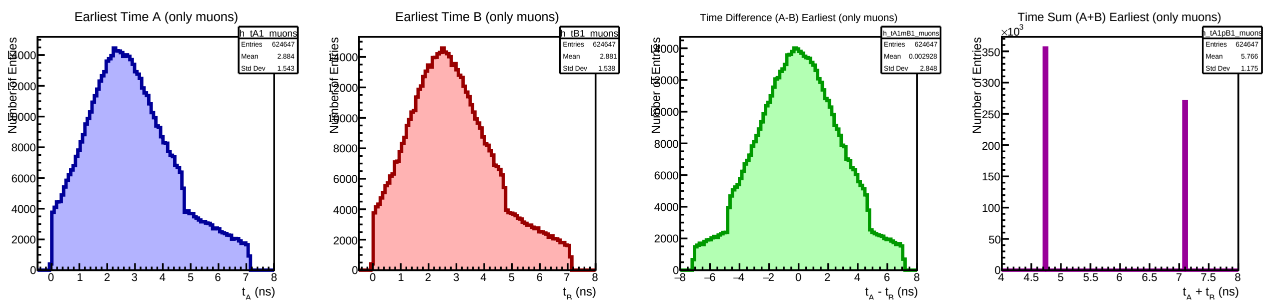


Figure 14: Temporal distributions for the earliest impact in the scintillating bars.

To demonstrate that the previous distribution is indistinguishable from those obtained with the latest impacts, we will plot the difference between the two distributions:

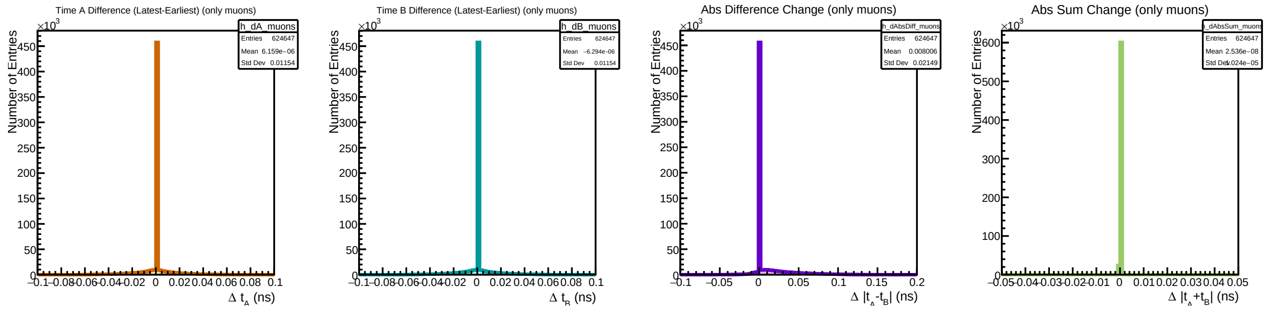


Figure 15: Distributions of the temporal distributions differences between earliest and latest impacts.

What we observe from the previous graphs is that these distributions can be approximated by a Dirac delta function. Therefore, we can use only one of the hits to perform the analysis (in this case we have chosen the earliest one). The distributions have a relatively low spread, and as we have already seen, only $\sim 9\%$ of the muons leave more than one hit in each scintillating bar.

Now, in order to study the uncertainties, we need to assign some uncertainty values to the previously presented temporal values. To achieve this, the procedure will be to apply some smearing to this values of a certain value, σ_t, that will be defined as the uncertainty of the temporal variables.

This involves applying a random gaussian distribution with a mean of 0 and a width of σ_t, resulting in new values of $t_{A,Smeared}$ and $t_{B,Smeared}$. These values can then be used to reconstruct the impact coordinates of the cosmic ray.

In order to reconstruct the point, we must consider that the plane 1 and plane 2 bars are perpendicular to each other, and that the time values only allow us to recreate the coordinate along which the scintillating bar is aligned. For plane 1, we obtain a value of z_{Reco} from the values t_A and t_B and for x_{Reco} and y_{Reco} we assign the middle point of the bar (which, for the local reference frame is 0, however, for the global reference frame is the middle point of the bar with respect to this system). For the plane 2 we do the same, but in this case the variable reconstructed from the times is x_{Reco}.

If instead of using t_A and t_B, we use $t_{A,Smeared}$ and $t_{B,Smeared}$, there will be some uncertainties in each coordinate. For the cases where the coordinate is set as the middle point of the bar, we will assign an uncertainty of:

$$\sigma_{geom} = \frac{bar\_width}{\sqrt{12}} = \frac{15}{\sqrt{12}} = 4.3\ mm, \tag{2}$$

where we have considered a square distribution.

And for the case when we reconstruct the coordinate from the temporal values, with the assigned group velocity, we can obtain the uncertainty by using the propagation of uncertainties formula:

$$\sigma_{reco} = \frac{\sigma_t\ v_g}{\sqrt{2}} \tag{3}$$

We will now present the distributions of the reconstructed coordinates, both with and without smearing the times. To achieve this, we will plot the difference between the reconstructed coordinates and the original coordinates saved from the Geant4 simulation.

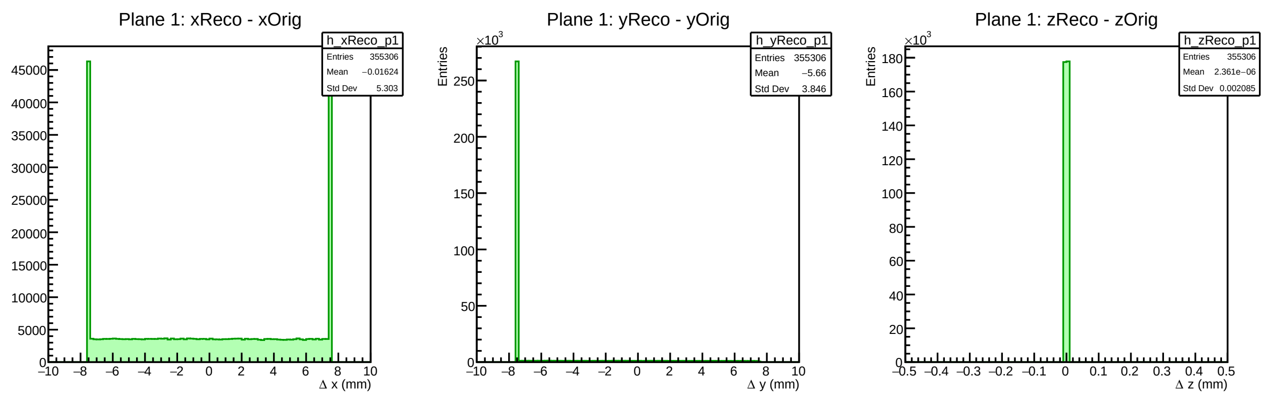


Figure 16: Distribution of the reconstructed coordinates for plane 1.

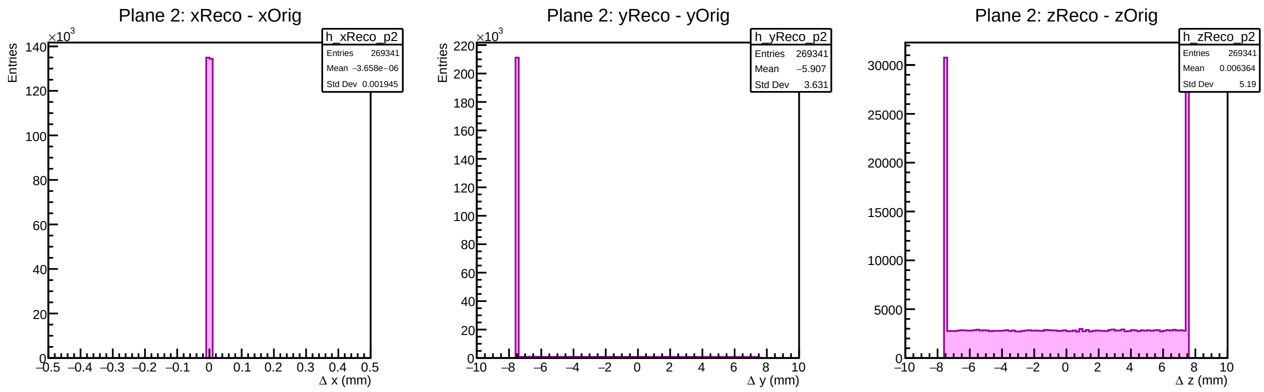


Figure 17: Distribution of the reconstructed coordinates for plane 2.

When we apply smearing to the temporal values, in this case with a value of $\sigma_t = 0,1\ ns$, the result is as follows:

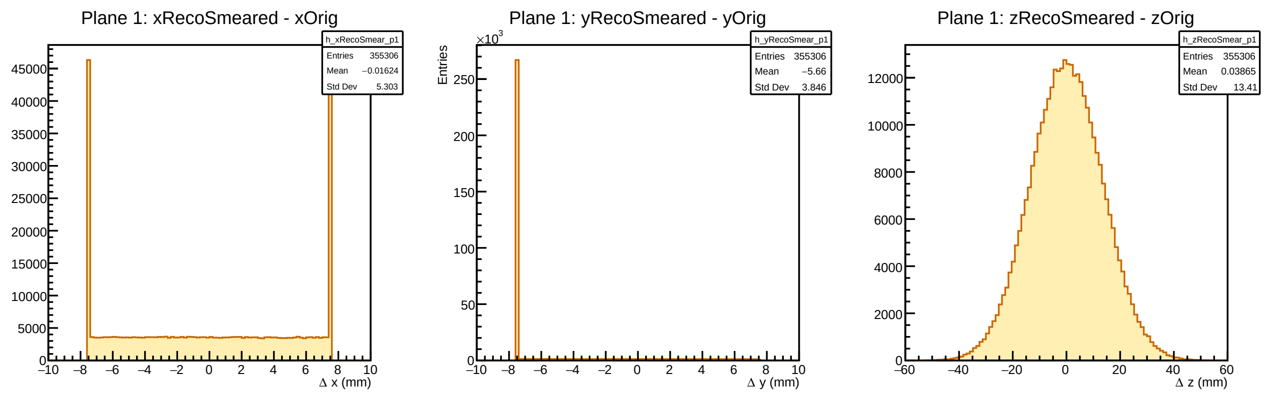


Figure 18: Distribution of the reconstructed coordinates for plane 1 applying some smearing to the time values.

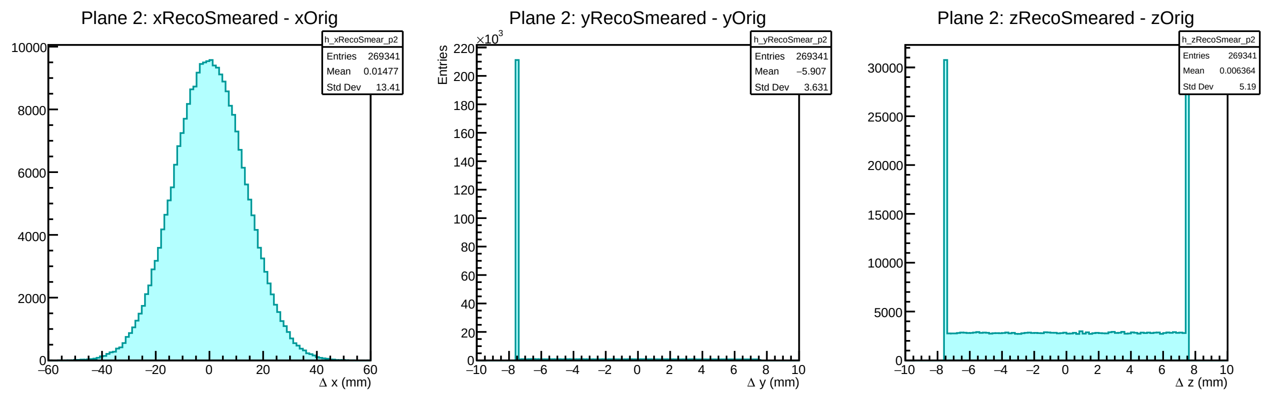


Figure 19: Distribution of the reconstructed coordinates for plane 2 applying some smearing to the time values.

These graphs allow us to make some observations. Firstly, the reconstruction procedure works perfectly well, as expected, since we are simply reversing the parameterisation performed in the Geant4 simulation.

Looking at the disrtibutions of the y coordinate, we can see that assigning the middle point in this case is not correct, since the $yOrig$ is $\sim 8\ mm \sim \dfrac{bar\_width}{2}$ above the reconstructed coordinate in all cases. It seems more appropriate to set the upper value of y for each plane and not consider any uncertainty for this coordinate, since we are again seeing a Dirac delta function.

For the coordinates that are not reconstructed from the time values (x for plane 1 and z for plane 2), we can observe a square distribution of the values, with higher values at the extremes, as we have already noticed in Figure 13, since there are more impacts at the extremes of the bars

Lastly, for the coordinates reconstructed with the temporal values after the smearing occurred, we observe a gaussian distribution of the values with some width. This width is the important parameter for us, since it should correspond to the uncertainty presented in equation 3.

Thus, for several time smearings we can study the uncertainty of the coordinates by pergorming a gaussian fit on the previous graph. The following graph presents the values of σ_z and σ_x obtained with the gaussian fit and the theoretically expected uncertainty from equation 3

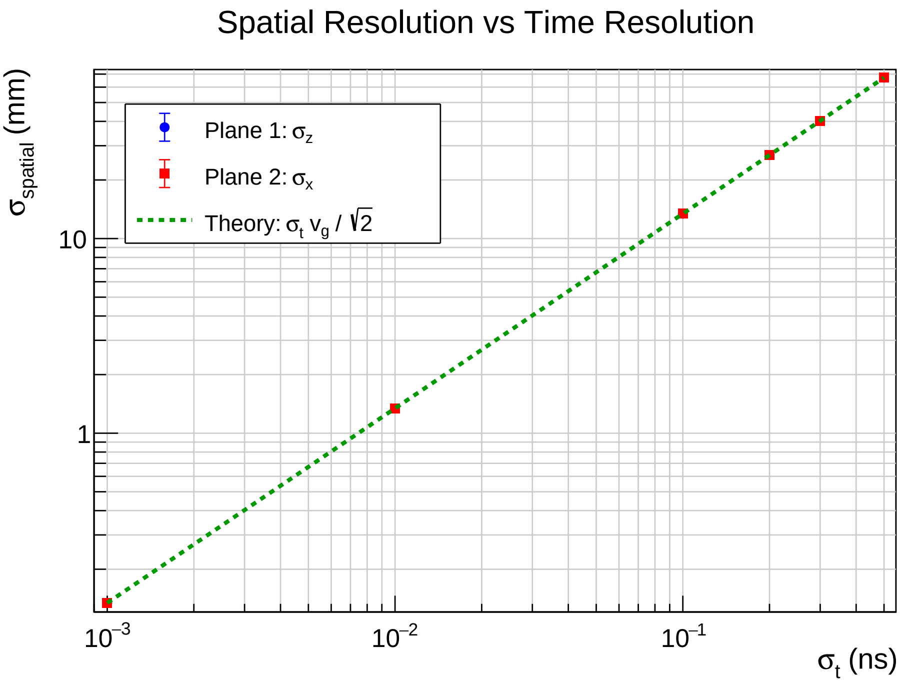


Figure 20: Spatial resolution for different temporal resolutions (time smearings).

Here, we can see that a time resolution grater than 0.1 *ns* would imply a spatial resolution of around ~ 10 *mm*, which is significant, considering that the QPU is 22 *mm* long. However, the important spatial uncertainty is that of the projected point in the QPU.

In oder to reconstruct the trajectory of the cosmic ray, we need to associate for each event a point for plane 1 and another point for plane 2. To do this, we need to consider that some rays may pass through more than one bar in each plane.

To study this, we start by analysing the pattern of the hits associated with each event. In principle, we would expect the most common options to be either one hit per plane, one hit in one of the planes and two hits in the other, or even two hits per plane. We present the distribution of the patterns associated with each event according to this classification:

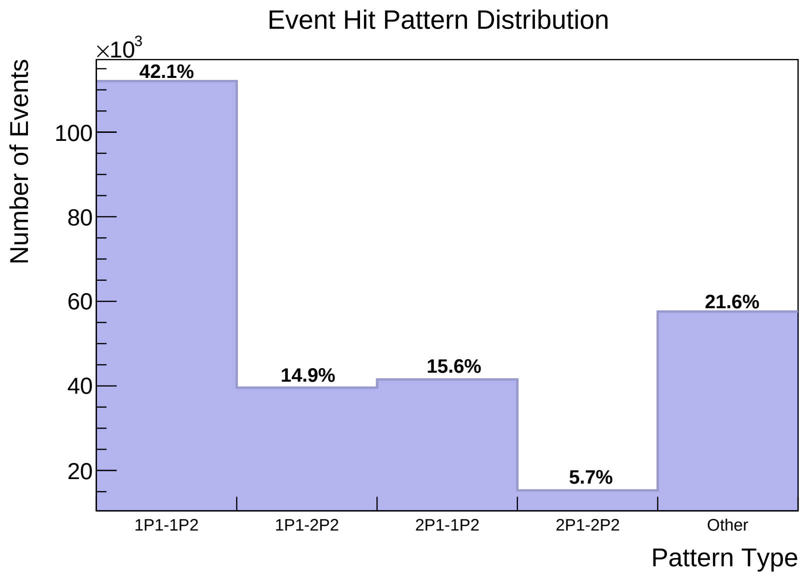


Figure 21: Expected patterns in the cosmic rays trajectories.

We can see that there are many unexpected events. To study what is happening here, we perform a more detailed analysis and observe the following:

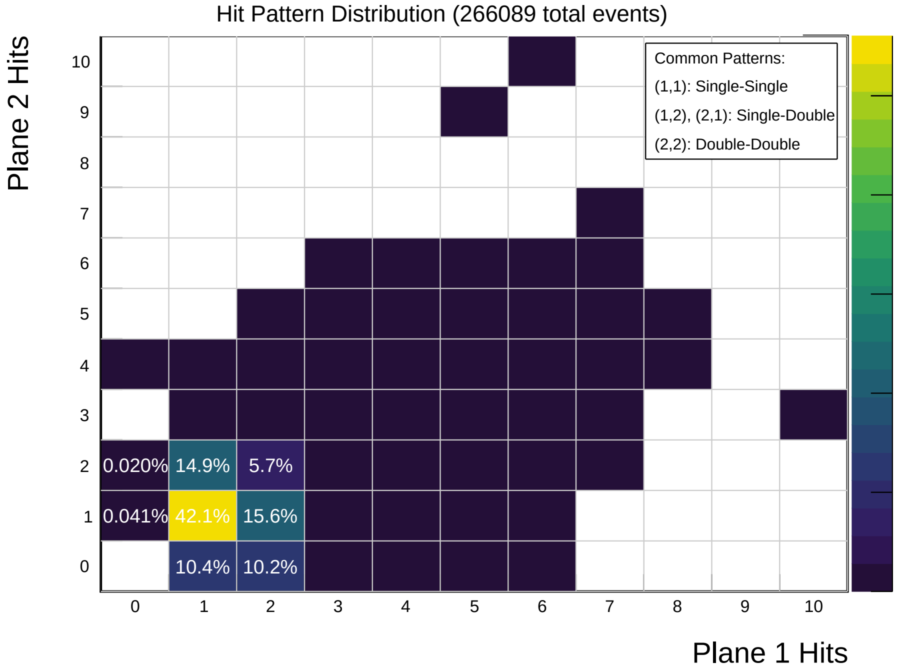


Figure 22: All measured patterns in the cosmic rays trajectories.

Some of the options with 10 hits per plane suggest there may be some unexpected behaviour occurring in the simulation. However, these behaviours are only present in one of the 270000 events. Assuming this can be omitted, What we can observe is that what we identified in the previous graph as 'other patterns' is mainly a category with hits only in the plane 1, corresponding to 20.69% of the events.

We also studied the distribution of hits according to bar ID to identify any irregularities. We found the following:

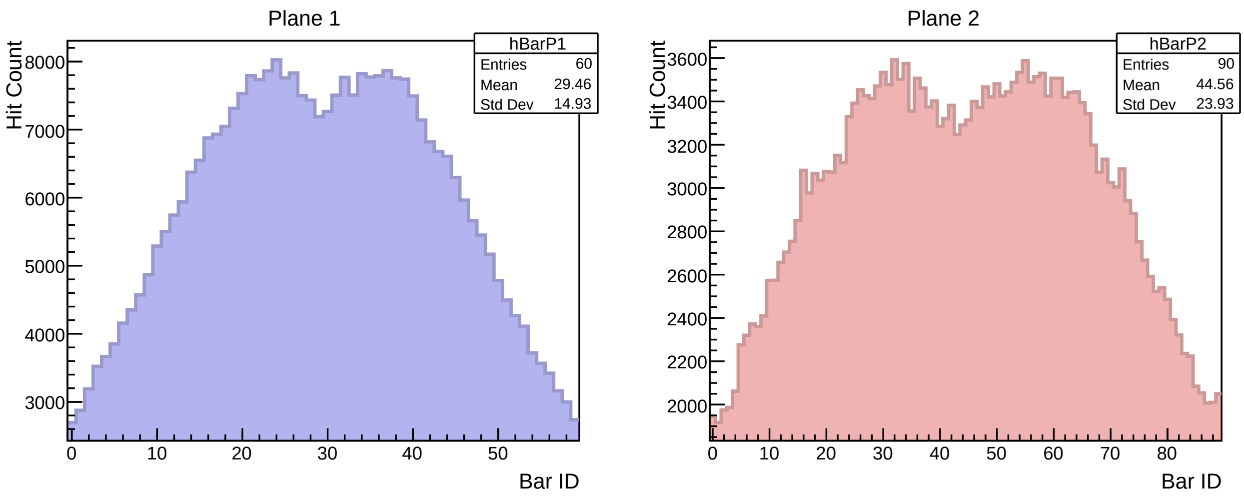


Figure 23: Distribution of the hits according to the bar ID.

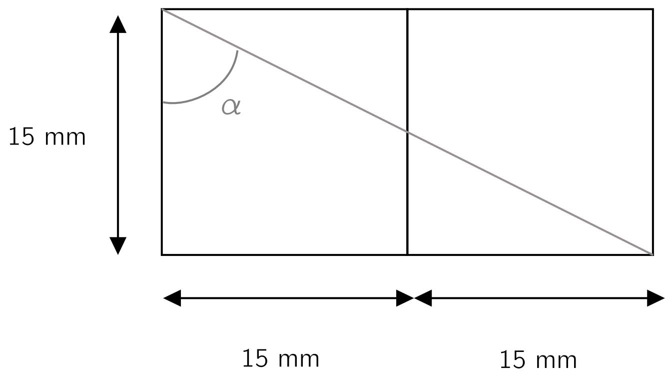
As can be seen for both planes, the distributions present two peaks instead of one, as could be inferred from the previously presented distributions of the impacts.

In order to continue with the analysis, we need to establish a method for processing cases where two hits are recorded on a given plane. The impact point for that plane could be set as the midpoint, but the associated uncertainty might not be trivial. In principle, we could expect an uncertainty of:

$$\sigma_{spatial} = \frac{2 * bar\_width}{\sqrt{12}} = \frac{30}{\sqrt{12}} = 8.6\ mm, \tag{4}$$

since we have a broader length covered by the cosmic ray trajectory in that plane, but this uncertainty can be reduced by taking into account the fact that we are only interested in the cosmic rays coming from the QPU.

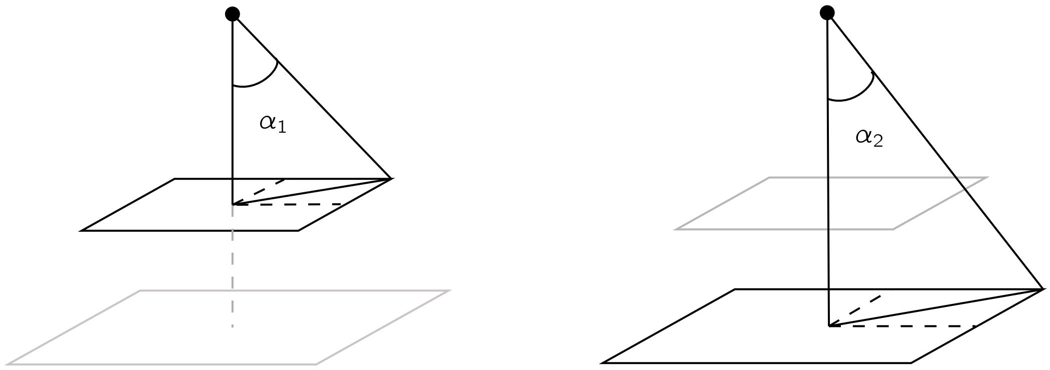
First, let's analyse the minimum value of an angle of incidence, α for a ray to pass through three different bars. This is the same value that a ray must have in order to be associated with the previously presented uncertainty.



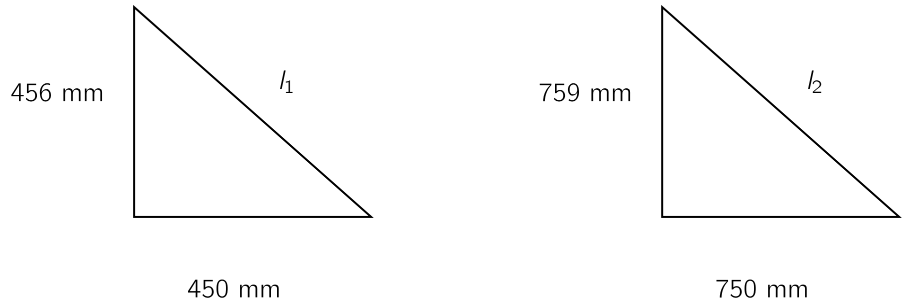
Let's compute this value:

$$tan\ \alpha = \frac{30}{15} = 2 \quad \Rightarrow \quad \alpha = arctan(2) = 63,47^{\circ}. \tag{5}$$

Now, we can take into account that the relevant trajectories are the ones that go through the QPU, that is, we will have a maximum α value for the cosmic rays we are interested in. The two following options are considered:



To calculate these values, we start by considering the triangle formed on the planes to find the hypotenuse. Here, we take into account the thickness of the coating.



The values obtained are:

$$l_1 = 640,65\ mm \qquad l_2 = 1067,04\ mm$$

And with this values we can obtain easily the values α_1 and α_2 taking into account that the QPU is at $y_{qpu} = 500\ mm$, the plane 1 at $y_1 = 0\ mm$ and the plane 2 at $y_2 = -500\ mm$. With this we obtain the values:

$$\alpha_1 = 52,03^\circ \qquad \alpha_2 = 46,86^\circ.$$

Now we have found that the relevant angle for as is α_2, since for angles greater than those, the plane 2 will not detect their presence, thus we will not be able to recreate the trajectory.

Using the same argument we used to compute the value of α, we can obtain the length covered by this new angle. Instead of 30 mm, now we obtain a value of:

$$tan\ \alpha_2 = \frac{d}{15} \quad \Rightarrow \quad d = 16,01\ mm,$$

and therefore we can assign an uncertainty to the point obtained as the mean between the impacts in two different bars of:

$$\sigma_{2p} = \frac{16,01}{\sqrt{12}} = 4,6\ mm \tag{6}$$

We note here that the previously computed uncertainty is valid for the coordinates that are not reconstructed from the time values, that is, the x coordinate for plane 1 and the z coordinate for plane 2. For the y coordinate, we have already established that we can set the values at the top of the planes without uncertainty. Lastly, for the case of the coordinates computed from the time values, considering the impacts in two bars will result in computing the mean of the values, and for the uncertainty we will use the uncertainty propagation rule for the middle-point formula.

Now, we have a method to assign to each plane a unique impact point $P_1 = (x_1, y_1, z_1)$ and $P_2 = (x_2, y_2, z_2)$.

After performing this computation, in which we reduce the number of impact points from two to one per plane, we obtain some distributions for the x and z coordinates in each plane.

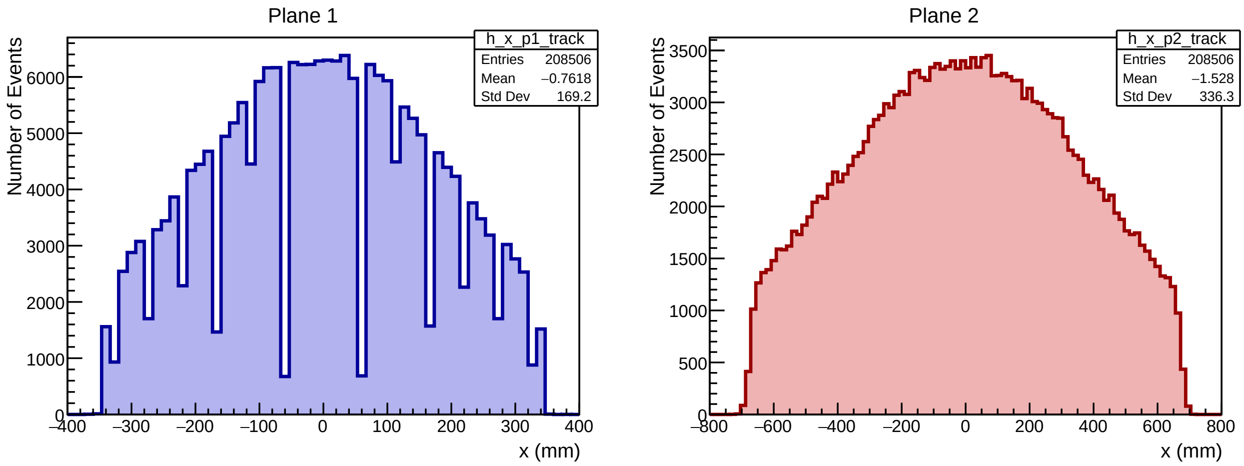


Figure 24: Histograms for the final impact x coordinates of both planes 1 and 2, after their treatment

For the coordinates reconstructed by the temporal values in each plane we see the expected distribution, however, when considering the other coordinates (omitting the y coordinate), we see that after performing the treatment of the data, that is, converting two hits per plane into one, some bars seem to be less populated for plane 1 and overpopulated for plane 2.

I

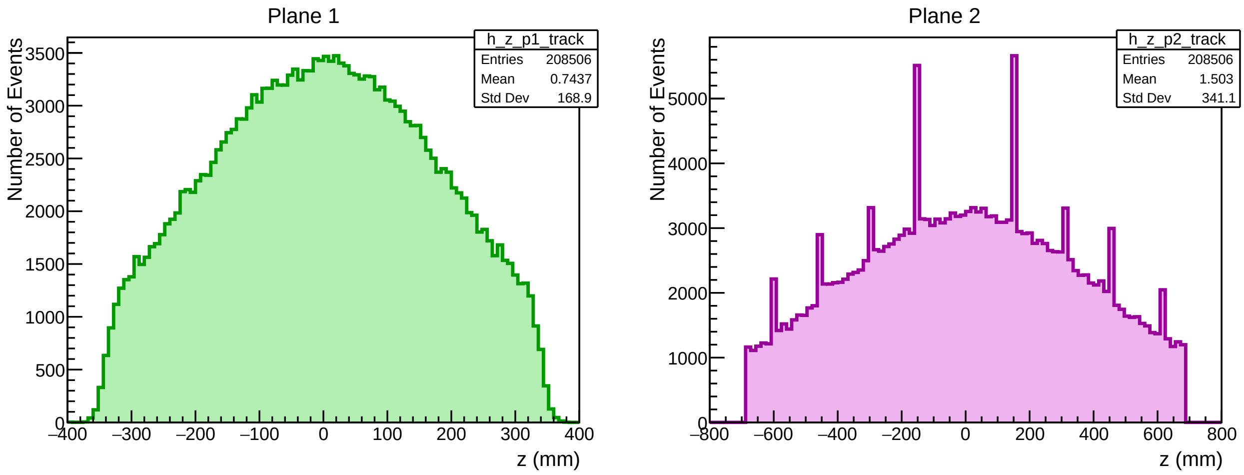


Figure 25: Histograms for the final impact z coordinates of both planes 1 and 2, after their treatment.

We can also plot a colour map of the treated points within the detector area.

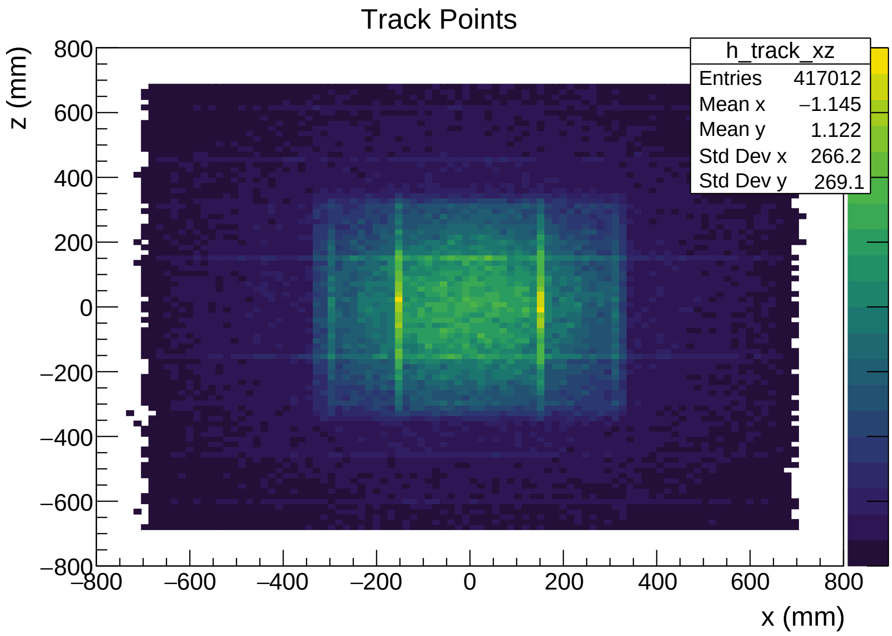


Figure 26: 2D histogram of the final considered impact points, after their treatment.

We can see that there are four or preferred bars in plane 1 and two in plane 2.

All the cases involving more than two hits per plane are discarded, as the angle at which they pass through the plane is too great for them to cross the QPU.

By defining the straight line that passes through these two points, we can reconstruct the cosmic ray track.

We can define a parametric line passing through two points P_1 and P_2 as:

$$P(t) = P_1 + t\,(P_2 - P_1). \tag{7}$$

The parameter t^* is chosen such that $y(t^*) = y_{\text{target}}$. Since:

$$y(t) = y_1 + t\,(y_2 - y_1), \tag{8}$$

we obtain:

$$t^* = \frac{y_{target} - y_1}{y_2 - y_1} = \frac{500 - 0}{-500 - 0} = -1. \tag{9}$$

The predicted point at $y = y_{target}$ is then:

$$P_{pred} = P_1 + t^*\,(P_2 - P_1) = 2P_1 - P_2. \tag{10}$$

Componentwise:

$$x_{\text{pred}} = x_1 + t^*(x_2 - x_1) = 2x_1 - x_2, \tag{11}$$
$$z_{\text{pred}} = z_1 + t^*(z_2 - z_1) = 2z_1 - z_2. \tag{12}$$

Using these two equations, we can determine the values of the coordinates at the QPU level, that is, for $y_{target} = y_{qpu} = 500\ mm$, and the uncertainties associated with the previous values are obtained using the standard uncertainty propagation procedure.

We obtain the following distributions of the previously presented coordinates, for a time resolution of $\sigma_t = 0,1\ ns$.

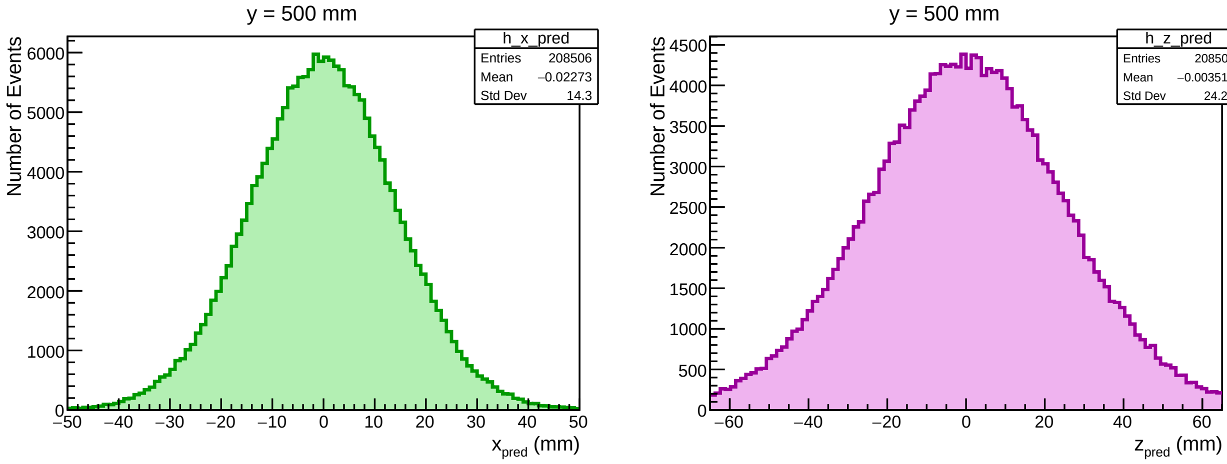


Figure 27: Histograms for the reconstructed x and z coordinates at the QPU level.

We can also represent the uncertainties distributions to gain an understanding of the values we can expect.

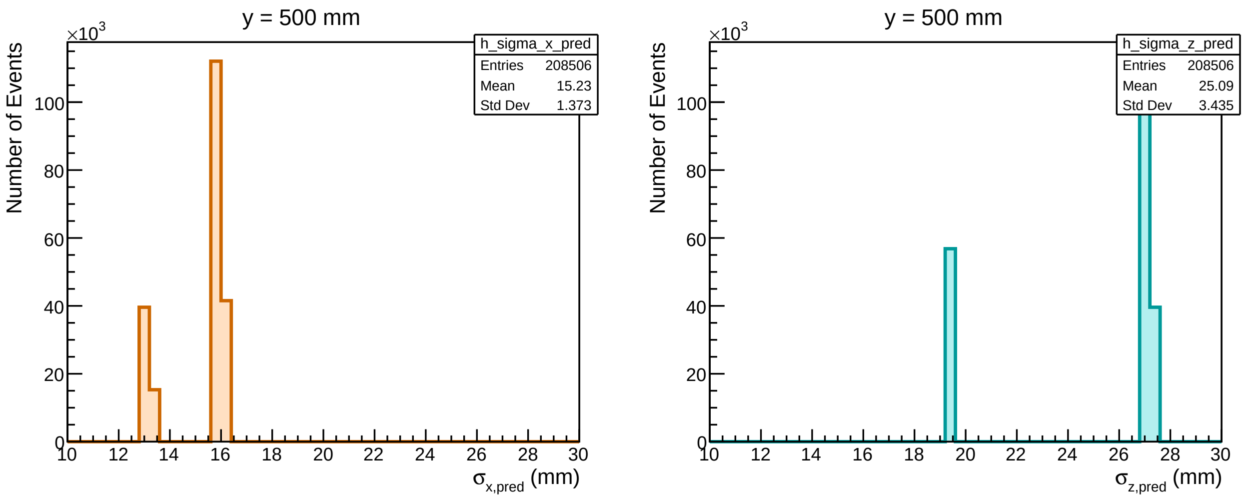


Figure 28: Distribution of the uncertainties for the predicted values of x and z at the QPU level.

We can also display the colour map of the reconstructed hits to illustrate the dispersion from the correct value, which is $(0, 500, 0)$ mm.

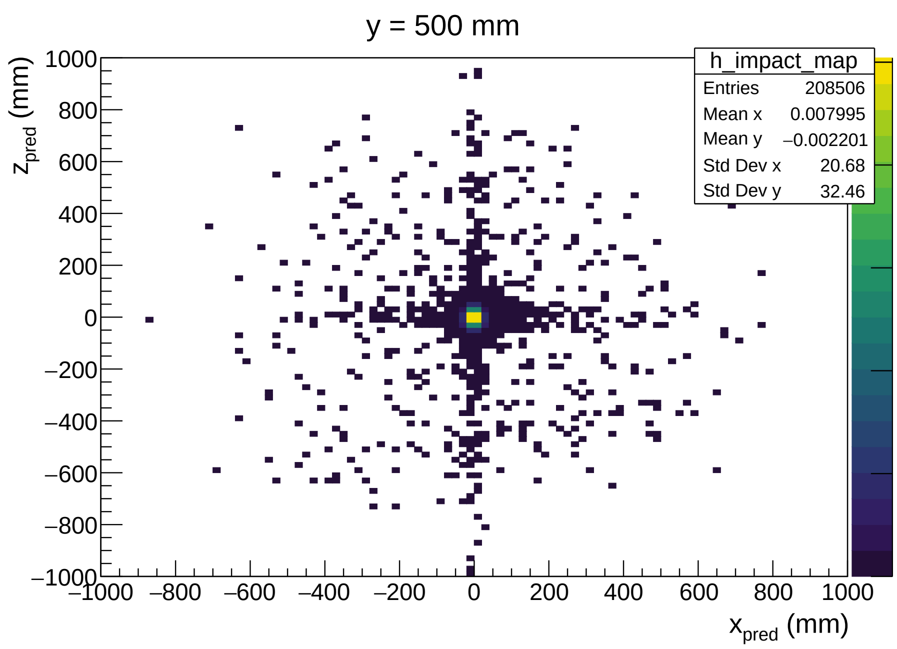


Figure 29: 2D histogram of the predicted values x and z at the QPU level.

As we can see, we have reconstructed some of the points far from where we expected.

Lastly, we can perform the same analysis with the uncertainties as we did for the reconstructed points. We can fit the distributions of the x_{pred} and z_{pred} variables (Figure 27) to a gaussian distribution to obtain the width parameter, which would be the parameter we will set as the uncertainty of the measurement.

After doing that for several time resolutions (σ_t), we obtain the following:

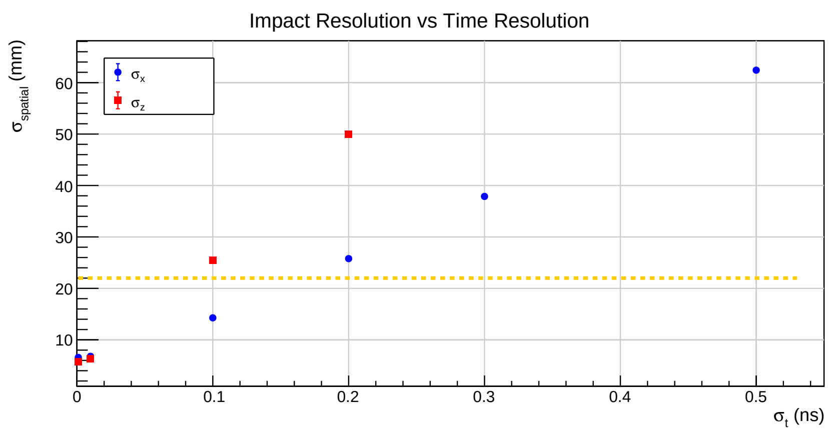


Figure 30: Spatial resolutions of the predicted coordinates for different values of the time resolution. The vertical orange line represents the value of 22 mm, which is the length of the QPU.

Explicitly:

σ_t (ns)	$\sigma_{x,pred}$ (mm)	$\sigma_{z,pred}$ (mm)
0.001	6.58(01)	5.72(01)
0.010	6.80(01)	6.32(01)
0.100	14.28(02)	25.46(04)
0.200	25.79(05)	49.93(09)
0.300	37.87(07)	74.55(13)
0.500	62.42(11)	124.07(22)

Table 1: Resolution in x_{pred} and z_{pred} as a function of σ_t.

We can see that for values grater than $0,1$ ns of time resolution we will not be able to discern whether the

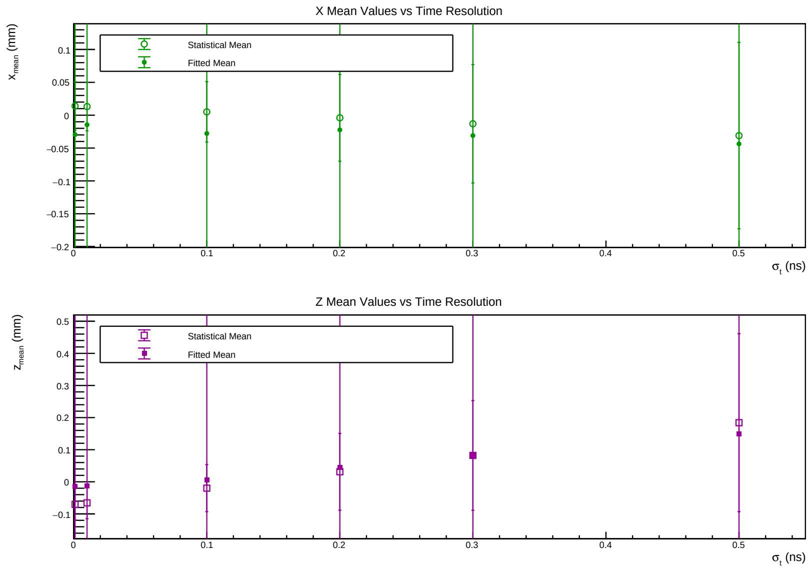


Figure 31: Mean values of the predicted variables as a function of the time resolution (σ_t).

cosmic ray passed through the QPU, since the uncertainty is bigger than the size of the QPU.

Lastly, for completeness, we will present the mean values of the reconstructed variables, although the relevant information is presented in the previous graph.

We observe that the mean values are all closer to 0, with small variations. However, we need to bear in mind that these values are averages. The statistical uncertainties (those inside the represented range) are obtained as the uncertainty of the mean and are therefore small. However, these values differ from those corresponding to the uncertainty of each predicted value, which are presented in the previous graph and are the purpose of the simulation.

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